

Topological Origins of the Dark Sector via K3 Compactification: The String Axiverse, UV Consistency, and Vacuum Energy

Xavier Callens¹

¹*SocrateAI Open Lab, Independent Research Node, France*

(Dated: July 14, 2026)

We propose a phenomenological ultraviolet (UV) embedding for Fuzzy Dark Matter (FDM) and Dark Energy by mapping the $S_{1,2}$ Picard-Fuchs effective field theory into a Type IIB string compactification on a K3-fibered Calabi-Yau threefold. By identifying the fundamental constituents of the dark sector with the exact topological invariants of a K3 surface, we construct a highly constrained framework. In alignment with recent extra-dimensional resonance models (Lee & Tsai, Phys. Rev. D, 2026), the integration of the Ramond-Ramond 4-form over the $b_2 = 22$ homology cycles naturally generates the String Axiverse. Here, the exact-rational $S_{1,2}$ sequence governs the Euclidean D3-brane instanton action, isolating a $\sim 10^{-21}$ eV Kaluza-Klein resonance suitable for FDM. Furthermore, the K3 Euler characteristic ($\chi = 24$) provides a necessary geometric condition for integer D3-brane tadpole cancellation (the explicit flux construction remains open), while driving the α' corrections of the Large Volume Scenario (LVS) that stabilize the internal volume; a metastable de Sitter uplift remains conditional on resolving the quintessence–Swampland tension kernel-verified in Part II. Finally, the severe chiral asymmetry dictated by the Hirzebruch signature ($\tau = -16$) yields a stress-energy Trace Anomaly via the Atiyah-Singer Index Theorem, providing a rigorous quantum field theoretic origin for the repulsive Dark Energy vacuum.

I. INTRODUCTION

The requirement that Effective Field Theories (EFTs) must couple consistently to quantum gravity places severe restrictions on low-energy phenomenology. Theories that fail these consistency conditions are relegated to the “Swampland”. In recent years, Fuzzy Dark Matter (FDM)—composed of ultra-light scalar fields with masses $m_a \sim 10^{-21}$ eV—has emerged as a leading candidate to resolve the small-scale crises of Cold Dark Matter.

In string phenomenology, the masses and couplings of scalar fields are rigidly determined by the topology and geometry of the internal compactified extra dimensions. In this work, we demonstrate that a Type IIB orientifold compactification on a K3-fibered Calabi-Yau threefold provides a mathematically motivated geometric origin for both the FDM scalar and the Dark Energy vacuum.

Remark 1 (Epistemic Scope). *This paper establishes a phenomenological mapping between K3 topological invariants and the dark sector. We do not claim to have constructed a fully stabilized string vacuum. The moduli values required to achieve explicit mass scales are treated as free parameters — indeed, the specific Kähler moduli used elsewhere in this series to reproduce the 10^{-21} eV targets are confirmed to have been fit to those targets rather than derived (Part I, Limitations, 2026-07-14 provenance resolution) — and exact tadpole flux vectors require future top-down derivation (an explicitly open problem seeking collaborators; *OPEN_PROBLEMS.md*, items 1–2). We further note that Part II of this series kernel-verifies a quintessence–Swampland obstruction ($\lambda_{\text{fit}} = 1.6724 > \sqrt{2}$) for the T^2 modulus; the de Sitter discussion in §?? below must be read as conditional on resolving that tension, not as an established vacuum construction.*

II. TOPOLOGICAL ORIGINS OF THE DARK SECTOR

A. Defining the Geometry

In algebraic geometry, a K3 surface is uniquely characterized by three exact integer invariants: the second Betti number $b_2 = 22$, the Euler characteristic $\chi = 24$, and the Hirzebruch signature $\tau = b_2^+ - b_2^- = 3 - 19 = -16$.

B. Generate Dark Matter (The Axiverse): $b_2 = 22$

Recent theoretical models, such as the 5D orbifold framework by Lee & Tsai [?], propose that dark matter is a natural mass-resonance enforced by the geometry of extra dimensions. Our 6D framework provides a concrete string-theoretic realization of this mechanism.

Because a K3 surface has exactly $b_2 = 22$ independent 2-cycles, integrating the Ramond-Ramond 4-form (C_4) field over K3 mathematically yields exactly 22 distinct axion fields:

$$a_i(x) = \int_{\Sigma_i} C_4, \quad i \in \{1, 2, \dots, 22\}. \quad (1)$$

The masses of these axions are generated by non-perturbative Euclidean D3-brane instantons wrapping the corresponding 4-cycles. The $S_{1,2}$ Picard-Fuchs sequence governs the specific topological rigidity of the primary cycle, defining the geometric resonance structure of the FDM axion; the 10^{-21} eV scale itself is reached only for Kähler moduli tuned within a two-parameter $(\tau, \mathcal{V})$ family (an achievability statement, not a unique prediction — see the Epistemic Scope remark and Part I, Limitations), isolating the FDM candidate from the remaining spectator axions.

C. A Necessary Condition for UV Consistency: Tadpole Structure ($\chi = 24$)

In F-theory and Type IIB orientifolds, the D3-brane charge must cancel against quantized fluxes to avoid fatal gauge anomalies, schematically:

$$N_{D3} + \frac{1}{2\kappa_{10}^2 T_3} \int_{\mathcal{M}} H_3 \wedge F_3 = \frac{\chi}{24}. \quad (2)$$

Because the Euler characteristic of the K3 fiber is strictly $\chi(K3) = 24$, the fiber contributes an exactly integer unit to the right-hand side — a *necessary* arithmetic condition for integer flux quantization that an arbitrary internal geometry would not satisfy. We emphasize this is not a demonstrated cancellation: the equation above is written for the relevant fourfold/orientifold data schematically, and exhibiting explicit flux quanta (F_3, H_3) with verified tadpole and anomaly cancellation for a concrete $K3 \times T^2$ orientifold remains an open problem for which we are actively seeking string-phenomenology collaborators (OPEN_PROBLEMS.md, item 1; task T7.2).

D. Toward Dark Energy: $\chi = 24$ and $\tau = -16$

In the Large Volume Scenario (LVS), the non-zero Euler characteristic ($\chi = 24$) generates the pertur-

bative α'^3 quantum corrections to the Kähler potential. This stabilizes the volume of the extra dimensions; obtaining a metastable de Sitter minimum from the generic LVS AdS vacuum additionally requires an uplift mechanism, and Part II of this series kernel-verifies that the simplest rolling- T^2 realization sits in the Agrawal–Obied–Steinhardt–Vafa quintessence–Swampland tension ($\lambda_{\text{fit}} > \sqrt{2}$). The de Sitter statement here is therefore a target of the programme, conditional on one of the three resolution paths identified in Part II, not an established result.

Furthermore, the Hirzebruch signature dictates the chiral asymmetry of the vacuum via the Atiyah–Singer Index Theorem ($n_+ - n_- = \tau = -16$). This severe chiral asymmetry prevents the perfect cancellation of vacuum fluctuations, sourcing the Trace Anomaly of the stress-energy tensor ($\langle T^\mu_\mu \rangle \neq 0$). This acts as an irreducible zero-point vacuum energy, establishing a geometric mechanism for a repulsive Dark Energy vacuum.

-
- [1] T. Lee and Y.-D. Tsai, *Naturally resonant dark matter from extra dimensions*, Phys. Rev. D **114**, L011701 (2026).